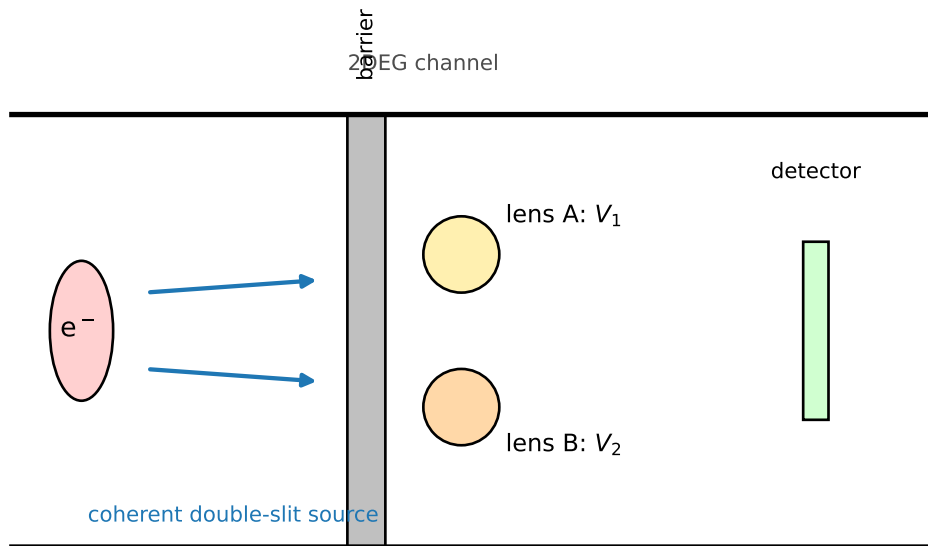


Scalable two-input XOR gate with independent electrostatic lenses

(a) Dual-lens interferometer



$I > I_{th} \Rightarrow$ logic 1

$I < I_{th} \Rightarrow$ logic 0

(b) XOR truth table (shallow-lens regime)

V_1	V_2	out
0	0	balanced (reset) 0 / 1
0	1	lower lens on 0
1	0	upper lens on 1
1	1	balanced (reset) 0 / 1

$\Delta\phi = \phi(V_1) - \phi(V_2)$ linearises the XOR truth table (Eq. 13)